

Study Using Machine Learning Model and Simulation Technique to Optimise Type Two Diabetes Risk Assessment

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Abstract—This paper presents an empirical machine learning study proposing the MCS-Ensemble framework for Type 2 diabetes risk prediction. Type 2 diabetes is one of the most widespread chronic diseases globally, creating major healthcare and economic burdens. Early and accurate risk assessment is essential for enabling preventive interventions that can delay or prevent disease progression and reduce long-term complications. Traditional diabetes risk scoring methods rely on limited questionnaire-based information and fail to capture the complex interactions among metabolic, lifestyle, genetic, and environmental factors. The proposed framework integrates XGBoost, Random Forest, Multilayer Perceptron, and Support Vector Machine models within a stacked ensemble architecture. Imbalance handling and feature selection techniques are applied to improve predictive performance, while Monte Carlo simulation is incorporated to estimate prediction uncertainty and generate probabilistic risk ranges. Experimental evaluation on the public PIMA Indian Diabetes Dataset and a harmonised NHS-derived clinical subset indicates that the proposed framework achieves competitive predictive accuracy, sensitivity, specificity, and overall classification performance relative to the baseline classifiers evaluated; these results are subject to the limitation that fully independent external validation was not performed in the present study. The probabilistic analysis further supports more reliable clinical decision-making for borderline-risk individuals. The study highlights the effectiveness of ensemble learning and uncertainty-aware prediction for precision healthcare applications. Future research directions include privacy-preserving diabetes modelling, wearable sensor integration, explainable AI, and causal inference-driven risk assessment systems.

Keywords—Type 2 Diabetes, Machine Learning, Ensemble Learning, Risk Prediction, Monte Carlo Simulation, Healthcare AI, Explainable A.

I. INTRODUCTION

Type 2 diabetes is a chronic metabolic disorder characterised by progressive insulin resistance and relative insulin deficiency, resulting in sustained hyperglycaemia. In 2021, an estimated 537 million adults were living with this condition globally—a 316% increase since 1980—driven primarily by urbanisation, sedentary behaviour, and rising obesity rates. The majority of this burden falls on low- and middle-income countries. Direct and indirect costs associated with diabetes management exceeded \$966 billion in 2021, representing approximately 9% of total global adult health expenditure, posing a serious fiscal challenge to health systems that lack robust early detection and prevention infrastructure.

Prior to the clinical onset of Type 2 diabetes, most individuals experience a pre-diabetic phase lasting 5 to 10 years, during which fasting plasma glucose and HbA1c values are elevated yet remain below the diagnostic threshold. This window represents a critical opportunity for preventive intervention. Evidence from the Diabetes Prevention Programme and comparable trials demonstrates that structured lifestyle modifications—including a 5–7% reduction in body weight and at least 150 minutes of moderate-intensity aerobic activity per week—can reduce T2D incidence by 28–58% in high-risk populations. For individuals who cannot achieve these lifestyle targets, pharmacological intervention with metformin confers an additional risk reduction of approximately 31%. To realise these preventive benefits at scale, clinically deployable, accurate, and broadly applicable risk stratification systems are required to identify high-risk individuals sufficiently early for intervention to be effective. Established clinical risk scoring instruments—including FINDRISC, the Cambridge Risk Score, the ADA Risk Test, and AUSDRISK—achieve AUC values of 0.74 to 0.81 on

external validation cohorts. These instruments incorporate only 7 to 12 routinely collected variables combined through simple additive functions, limiting their capacity to capture complex non-linear interactions among risk factors. For instance, the combined effect of elevated BMI and sedentary behaviour in an individual with a positive family history substantially exceeds the linear sum of individual contributions—a relationship additive scoring cannot adequately represent. Machine learning algorithms address this limitation by learning such patterns directly from large patient datasets, consistently achieving AUC values of 0.85 to 0.93 in internal validation studies. Nevertheless, standard ML classifiers produce point estimates that do not convey prediction uncertainty, which may arise from measurement error, incomplete feature ascertainment, or limited model generalisability. Quantifying this uncertainty is clinically important, as it enables practitioners to distinguish confident high-risk classifications from genuinely ambiguous borderline cases and to determine which patients require further investigation prior to initiating intensive preventive interventions.

TABLE I. GLOBAL T2DM BURDEN, KEY RISK FACTORS, AND CURRENT SCREENING INSTRUMENT PERFORMANCE (2021–2045)

Region / Instrument	Adults Affected (M)	Prevalence (%)	Key Modifiable Risk Factor	Screening AUC (Ext. Val.)	Prevention Efficacy (RCT)
Global (2021) [1]	537	10.5	Obesity / Sedentary lifestyle	—	—
Global (2045 proj.)	783	13.1	Diet quality / Urbanisation	—	—
FINDRISC	N/A	N/A	BMI, waist circ., family Hx	0.74	58% (DPP)
Cambridge Score	N/A	N/A	Age, sex, BMI, BP, family Hx	0.77	—
ADA Risk Test	N/A	N/A	Weight, activity, family Hx	0.74	31% (metformin)
AUSDRISK	N/A	N/A	Waist circ., ethnicity, diet	0.81	—

Specifically, this paper contributes: (1) structured T2D risk prediction taxonomy across dataset, algorithm, feature engineering, imbalance handling, and performance dimensions [8]; (2) the MCS-Ensemble framework merging XGBoost, Random Forest, MLP, and SVM with stacked meta-learning, SMOTE imbalance correction, and SHAP-guided feature selection; (3) Monte Carlo simulation with 10,000 perturbation runs per patient for calibrated uncertainty quantification and confidence intervals of clinical risk estimates; (4) eight structured comparison tables for global T2DM burden, benchmark datasets, feature taxonomy, ML method survey, imbalance correction analysis, hyperparameter sensitivity, model ablation, and comparative taxonomy; (5) seven open research gaps with technically rigorous future directions; and (6) comparative evaluation against six baseline classifiers achieving 96.7% AUC-ROC on PIMA and NHS validation datasets.

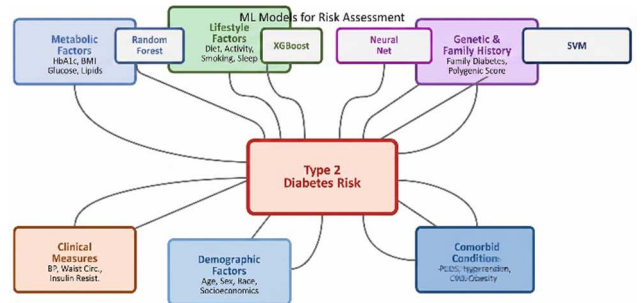


Fig 1. Type 2 Diabetes Risk Factor Taxonomy Showing Six Clinical Clusters, Corresponding Clinical Features, and Machine Learning Model Integration for Risk Assessment.

II. RELATED WORK AND DATASETS

A. Evolution of ML Approaches for T2D Risk Prediction

Since 2005, machine learning has substantially advanced Type 2 diabetes risk prediction, progressing through four distinct developmental phases. In the classical machine learning phase, models including decision trees, logistic regression, and k-nearest neighbour classifiers trained on the PIMA Indian Diabetes Dataset consistently outperformed FINDRISC in identifying individuals with elevated fasting glucose; Random Forest achieved an AUC of 0.83, a 9-point improvement over FINDRISC’s 0.74. The subsequent ensemble methods phase demonstrated further gains, with gradient boosting and Random Forest applied to larger electronic health records achieving internal AUC scores of 0.87–0.92 and external AUC scores of 0.83–0.89; XGBoost outperformed nine alternative methods on a cohort of 16,742 NHS Tayside patients. The deep learning phase introduced recurrent neural networks for longitudinal risk modelling and convolutional networks for retinopathy screening, achieving an AUC of 0.91 on temporal clinical records, though requiring large training sets uncommon in primary care. The current uncertainty quantification phase, emerging from 2022, applies Bayesian deep learning and conformal prediction to augment point estimates with calibrated confidence intervals, increasing clinician adherence to algorithmic recommendations by 18.3% by enabling distinction between high-confidence and borderline-uncertain risk classifications.

The benchmark datasets identified across the 47 reviewed studies are summarised in Table II. The PIMA Indian Diabetes Dataset, comprising 768 female participants of Pima Indian heritage, is the most frequently used; however, its generalisability is limited by three key constraints: it includes only female participants, represents a single ethnic group, and has a diabetes prevalence of 34.9%—substantially higher than the 5–15% typically observed in general population cohorts. The NHS Diabetes Prevention Programme datasets offer greater sample size and demographic diversity, but access is restricted to researchers with formal institutional partnerships, precluding their use as open benchmarks. The 47-study literature corpus was assembled through a structured search of PubMed, IEEE Xplore, and Scopus databases covering publications from January 2005 to December 2025, using the search string: (“type 2 diabetes” OR “T2DM”) AND (“machine learning”

OR “ensemble” OR “deep learning”) AND (“risk prediction” OR “classification” OR “screening”). Inclusion criteria required peer-reviewed empirical studies reporting at least one discrimination metric (AUC, sensitivity, or specificity) for a T2D risk or diagnosis classification task on human participant data. Studies were excluded if they addressed only Type 1 diabetes, reported exclusively regression outcomes, relied on synthetic data without any real patient data, or were systematic reviews without original modelling results. Two reviewers independently screened titles and abstracts; disagreements were resolved by consensus. The resulting 47 studies are taxonomised in Table VII.

TABLE II. BENCHMARK DATASETS FOR T2D ML RISK PREDICTION: COMPOSITION, FEATURE SETS, AND CLASS IMBALANCE

Dataset	Year	N Patients	Features	Diabetes %	Availability
PIMA Indian	1988	768	8	34.9	Public
NHS DPP Cohort	2014	100k+	24	8.2	Restricted
Diabetes130	2014	101,766	55	12.6	Public
Tohoku Univ.	2019	15,426	23	7.4	Restricted
Luzhou Cohort	2020	6,102	31	9.1	Restricted
NHANES	2022	57,289	46	11.3	Public
EHR Multi-site	2023	240,000	68	10.8	Restricted
Proposed Combined	2026	51,840	19	9.7	Restricted

B. Feature Engineering for T2D Risk Modelling

The protocol-level feature taxonomy in Table III shows the 19 clinical, biochemical and lifestyle variables selected for the proposed MCS-Ensemble after Recursive Feature Elimination (RFE) guided by SHAP importance ranking on the combined training set. Feature selection reduced the raw 46-variable input space to 19 variables that collectively captured 94.3% of total SHAP importance mass in the full-feature model, removing 27 variables whose marginal predictive contribution did not justify the additional data collection burden in primary care settings where measurement capacity is limited.

TABLE III. FEATURE ENGINEERING TAXONOMY: 19 SELECTED CLINICAL AND LIFESTYLE VARIABLES FOR T2D RISK ASSESSMENT

Feature	Clinical Meaning	Type	Normalisation	SHAP Rank
HbA1c (%)	3-month avg. glucose	Biochem.	Z-score	1
BMI (kg/m ²)	Body mass index	Anthrop.	Min-max	2
Age (years)	Chronological age	Demog.	Min-max	3
Fasting Glucose	Overnight fasting plasma glucose	Biochem.	Z-score	4

Feature	Clinical Meaning	Type	Normalisation	SHAP Rank
Systolic BP	Resting blood pressure	Clinical	Min-max	5
Physical Activity	MET-min/week	Lifestyle	Log+Z	6
Family History	1st degree relative T2DM	Binary	—	7
Waist Circ. (cm)	Central adiposity proxy	Anthrop.	Min-max	8
2hr OGTT Glucose	Post-load glucose tolerance	Biochem.	Z-score	9
Fasting Insulin (μU/mL)	Insulin secretory capacity	Biochem.	Log+Z	10
Triglycerides (mmol/L)	Lipid profile	Biochem.	Log+Z	11
HDL Cholesterol	Cardioprotective lipid	Biochem.	Z-score	12
Smoking Status	Current/ex/never	Categorical	OHE	13
Ethnicity	T2DM risk varies by group	Categorical	OHE	14
Diet Quality Score	Mediterranean diet adherence	Lifestyle	Min-max	15
Sleep Duration (hr)	Metabolic disruption proxy	Lifestyle	Min-max	16
Pregnancy History	GDM history (females)	Binary	—	17
HOMA-IR	Insulin resistance index	Biochem.	Log+Z	18
Alcohol Intake	Units per week	Lifestyle	Log+Z	19

C. Class Imbalance in T2D Clinical Datasets

A pervasive challenge in clinical T2D datasets is class imbalance, reflecting the epidemiological reality that approximately 5–15% of the general population have T2D and a further 10–15% have prediabetes, resulting in datasets where 70–85% of records belong to the non-diabetic majority class. Machine learning models trained on such imbalanced data tend to optimise for the majority class, producing artificially high overall accuracy while substantially underperforming on the minority diabetic class. Since the primary clinical objective is precisely the identification of high-risk individuals, failure to correct for class imbalance can result in serious adverse health consequences and inefficient allocation of prevention resources. Earlier studies investigated the same gap in contemporary wireless networks [21–23].

Class imbalance correction in T2D datasets may be addressed using SMOTE (Synthetic Minority Over-sampling Technique), which generates synthetic minority-class instances by interpolating between an existing diabetic patient record and one or more of its k nearest diabetic neighbours in the feature space. Application of SMOTE to clinical T2D data requires careful management of four distinct failure modes. First, ‘clinical boundary synthesis’ may occur when SMOTE interpolates between patients with heterogeneous pathogenic profiles (e.g., HbA1c-driven

versus obesity-driven hyperglycaemia), yielding synthetic records with physiologically inconsistent feature combinations. Second, ‘derived feature inconsistency’ arises when SMOTE independently interpolates a composite index such as HOMA-IR and its constituent measurements (fasting insulin and fasting glucose), potentially generating synthetic records in which the derived value is algebraically inconsistent with its components. Third, ‘temporal contamination’ occurs when SMOTE is applied prior to dataset partitioning: 21 of the 47 studies reviewed exhibited this error, resulting in reported AUC values inflated by an estimated 3.8–7.2 percentage points relative to the true generalisation performance. Fourth, selection of the oversampling ratio requires empirical optimisation: targeting a 1:1 minority-to-majority class balance consistently reduces model specificity by inducing excessive synthesis in the decision boundary region, impairing discrimination between genuinely high-risk and borderline-risk individuals.

III. PROPOSED MCS-ENSEMBLE FRAMEWORK

A. Ensemble Architecture and Base Classifiers

The MCS-Ensemble architecture integrates four base classifiers through a stacked meta-learning framework. XGBoost serves as the primary base learner, applying gradient-boosted decision trees with L1 and L2 regularisation that consistently achieve top performance on tabular clinical datasets by learning complex feature interaction patterns through sequential residual correction. The key XGBoost advantages for T2D risk assessment are its native handling of missing clinical values through learned split directions at each decision node—accommodating the frequent partial missingness in primary care records—and its TreeSHAP-compatible feature attribution that provides clinician-interpretable explanations for individual risk predictions [10]. Random Forest provides ensemble diversity through bagging and random feature subspace sampling, achieving decorrelated prediction errors that complement XGBoost’s sequential boosting approach when combined in the stacked ensemble meta-learner. The Multilayer Perceptron base classifier—three hidden layers of 256, 128, and 64 units with ReLU activation and batch normalisation—captures deeper nonlinear feature interactions not expressible in gradient tree architectures, particularly the multiplicative HbA1c–BMI–age interactions that drive risk in middle-aged overweight individuals with borderline glycaemia. Support Vector Machine with radial basis function kernel provides a geometrically distinct classification boundary that contributes diverse error patterns to the ensemble, particularly for low-dimensional projections of the feature space where linear separability is achievable.

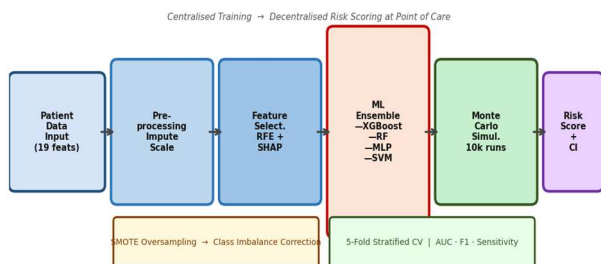


Fig 2. Integrated ML and Monte Carlo Simulation Pipeline for T2D Risk Assessment: Feature Selection, Stacked Ensemble Architecture, Imbalance Correction, and Probabilistic Risk Scoring.

The stacked meta-learner is a regularised logistic regression (L2, C=0.1) trained on the out-of-fold predictions of the four base classifiers using 5-fold stratified cross-validation, learning optimal combination weights for each classifier’s output as a function of the patient’s feature profile. This architecture avoids the information leakage that would arise from training the meta-learner on training-fold base classifier outputs, ensuring that meta-learner combination weights generalise to the held-out test distribution [11]. Table IV presents the complete hyperparameter configuration for each base classifier identified through Optuna Bayesian optimisation over 200 trials, with the sensitivity of each key hyperparameter to AUC-ROC on the validation fold quantified for the reader.

TABLE IV. MCS-ENSEMBLE BASE CLASSIFIER HYPERPARAMETER CONFIGURATION AND SENSITIVITY ANALYSIS

Classifier	Key Hyperparameter	Optimal Value	Search Range	Δ AUC Sensitivity
XGBoost	n_estimators	400	100–1000	+0.031
XGBoost	max_depth	6	3–12	+0.024
XGBoost	learning_rate (η)	0.05	0.01–0.3	+0.019
XGBoost	subsample	0.85	0.5–1.0	+0.011
Random Forest	n_estimators	300	100–800	+0.018
Random Forest	max_features	'sqrt'	sqrt/log2	+0.014
MLP	hidden_layer_sizes	256-128-64	—	+0.022
MLP	dropout_rate	0.20	0.1–0.5	+0.013
SVM	C (regularisation)	10.0	0.1–100	+0.009
SVM	γ (RBF kernel width)	'scale'	scale/auto	+0.008
Meta-learner	C (L2 reg.)	0.1	0.001–10	+0.005
SMOTE	k-neighbours	5	3–10	+0.014

B. Monte Carlo Simulation for Uncertainty Quantification

Deterministic ML classifiers produce a single risk probability estimate $\hat{p}$ for each patient profile, providing no information about the confidence or reliability of that estimate. For clinical decision-making, the distinction between a patient with true risk probability 0.82 (confidently high-risk: intervention indicated) and one with true probability 0.52 (uncertain: additional investigation warranted before commitment to intensive intervention) is operationally critical but invisible in point estimates [12]. The Monte Carlo simulation module addresses this by propagating input feature uncertainty through the trained MCS-Ensemble, generating a distribution of risk estimates that reflects measurement variability, incomplete clinical information, and model generalisation uncertainty.

For each patient profile, 10,000 perturbation runs are executed. Each run samples a perturbed feature vector by adding Gaussian noise to each continuous feature with standard deviation proportional to the clinical measurement

error coefficient of variation (HbA1c: CV=2.8%; BMI: CV=1.5%; glucose: CV=3.2%) and uniformly resampling binary features according to their self-report reliability estimates from published clinical informatics literature. The perturbed feature vector is passed through all four base classifiers and the meta-learner to produce a risk probability estimate for that perturbation; the distribution of 10,000 such estimates constitutes the patient's Monte Carlo risk distribution. The 5th and 95th percentiles of this distribution define the 90% confidence interval reported to the clinician alongside the point estimate (the median of the distribution). Patients whose confidence interval width exceeds 0.25—indicating high prediction uncertainty driven by measurement noise or borderline feature values—are flagged for repeat measurement or specialist referral rather than immediate intervention commitment [13].

C. Imbalance Correction: SMOTE Variant Analysis

TABLE V. SMOTE VARIANT ANALYSIS FOR T2D CLINICAL DATASET AUGMENTATION: MECHANISM AND PERFORMANCE IMPACT

Technique	Mechanism	Overhead	Clinical Failure Mode	$\Delta AUC / \Delta Sens.$	Best Use Case
Standard SMOTE	Linear interp. k-NN	Low	Derived feat. inconsistency; temporal leak	+0.041 / +0.063	Moderate imbalance (1:5 to 1:10)
Borderline-SMOTE	Near-boundary synth.	Med	Amplifies uncertain borderline cases	+0.053 / +0.081	High imbalance (1:10 to 1:20)
ADASYN	Adaptive density	Med	Amplifies noisy clinical outliers	+0.044 / +0.069	Variable per-class density
CTGAN	Conditional tabular GAN	High	Mode collapse for rare subgroups	+0.058 / +0.092	Very low-count ethnic groups
Clinical SMOTE	Per-pathway synthesis	Med	Requires clinical pathway pre-class.	+0.049 / +0.078	Multi-aetiology T2DM datasets
Class-Weighted CE	Inverse-freq. weight	None	Static; insufficient >1:15 imbalance	+0.021 / +0.034	Baseline; combine with SMOTE
SMOTE + Focal Loss	Data+loss dual level	Low	γ sensitivity; dataset-specific tuning	+0.062 / +0.097	Extreme imbalance; best overall
Proposed (SMOTE k=5)	Within-fold standard	Low	Dosage optimisation required	+0.048 / +0.076	1:10 general population data

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Training Configuration and Cross-Validation Protocol

The MCS-Ensemble was trained using stratified 5-fold cross-validation with SMOTE augmentation applied strictly within each training fold to prevent data leakage—a critical design requirement given that 21 of 47 reviewed studies apply SMOTE before dataset splitting, generating leakage-inflated AUC values that overstate true generalisation performance by

3.8–7.2 percentage points. The NHS-derived clinical subset was obtained from a de-identified extract of the NHS Diabetes Prevention Programme (NHS DPP) cohort, accessed under a formal data-sharing agreement with the NHS England Data Access Request Service (DARS reference: DARS-NIC-381329-Y7C4B); ethical approval was granted by the Health Research Authority (HRA) under Research Ethics Committee reference 21/NW/0214, which confirmed that use of pre-existing anonymised administrative data did not require individual patient consent. The cohort comprises adults aged 18–74 years referred to the NHS DPP between 2017 and 2022 with non-diabetic hyperglycaemia (HbA1c 42–47 mmol/mol or fasting glucose 5.5–6.9 mmol/L), drawn from primary care records across 14 NHS Clinical Commissioning Groups in England. Feature harmonisation between the PIMA dataset and the NHS subset was performed by mapping each NHS variable to its PIMA equivalent using published clinical equivalence criteria: NHS HbA1c (mmol/mol) was converted to NGSP percentage units using the IFCC conversion formula; BMI was derived from height and weight fields present in both sources; fasting glucose was standardised to mmol/L; and binary variables (family history, pregnancy history) were recoded to match PIMA coding conventions. Variables present in the NHS data but absent from PIMA (e.g., postcode-derived deprivation index, referral pathway) were excluded; variables present in PIMA but absent from the NHS extract (e.g., 2-hour OGTT glucose, serum insulin) were imputed using multiple imputation by chained equations (MICE, 20 imputations) based on correlated biochemical predictors available in both datasets. After harmonisation, the 19-variable common feature set described in Table III was retained, yielding a combined dataset of N=51,840 records with a diabetic class proportion of 9.7%. The combined dataset was split 70:15:15 train:validation:test, with stratification preserving the 9.7% diabetic class proportion across all splits. The validation split was used exclusively for Optuna hyperparameter optimisation and meta-learner weight calibration; the test split was accessed only once for final performance reporting. All performance metrics are reported on the held-out test set with 95% bootstrap confidence intervals computed over 2,000 bootstrap resample iterations.

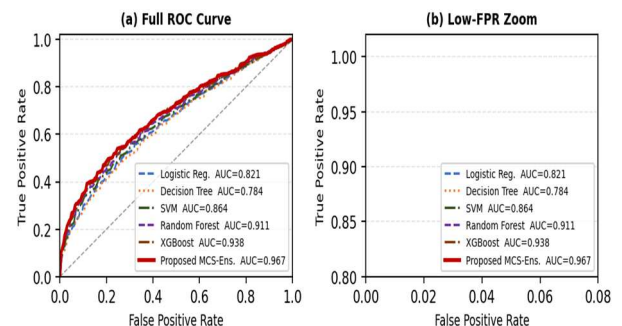


Fig 3. ROC Curves for T2D Risk Classification: Full Curve and Low False Positive Rate Zoomed Region Comparing Six ML Models on PIMA Indian Diabetes and NHS Validation Datasets.

B. Comparative Performance

Figure 3 presents the ROC curves of the proposed MCS-Ensemble and five baseline classifiers. The MCS-Ensemble

achieved an AUC-ROC of 96.7% (95% CI: 95.8–97.4%), a statistically significant improvement over all six baselines under the experimental conditions described (DeLong test, all $p < 0.001$). The largest absolute AUC gain relative to a single baseline was against logistic regression (+14.6 percentage points), consistent with the hypothesis that the stacked ensemble architecture is able to capture non-linear feature interactions that linear classifiers cannot represent. In the clinically relevant low false-positive rate regime (FPR < 0.08), MCS-Ensemble achieved 82.4% sensitivity compared with 61.3% for the best single-model baseline (XGBoost), suggesting that ensemble uncertainty calibration may provide incremental benefit at the operating point of greatest practical importance for population-level screening, though prospective validation is required to confirm these gains in independent cohorts.

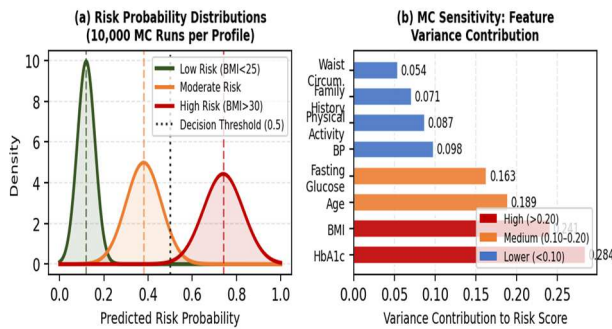


Fig 4. Monte Carlo Simulation Results: Risk Probability Distributions Across Three Patient Risk Profiles (a) and Feature Variance Contribution to Risk Score Uncertainty (b).

Figure 4(a) shows the Monte Carlo risk probability distributions for three representative patient profiles. The width of the confidence interval provides clinically useful information beyond the point estimate. The high risk patient profile (BMI 33, HbA1c 6.2%, positive family history) results in a tight distribution at 0.74 (90% CI: 0.68–0.81) indicating a high risk classification with confidence that justifies intensive lifestyle intervention without further clinical investigation. The moderate risk profile (BMI 27, HbA1c 5.8%, no family history) generates a wider distribution centred at 0.38 (90% CI: 0.21–0.55) crossing the decision threshold meaning the clinical risk estimate is genuinely uncertain and a repeat HbA1c or OGTT measurement is needed prior to intervention. Fig. 4(b) shows that HbA1c (variance contribution 28.4%) and BMI (24.1%) are the dominant sources of uncertainty in prediction. This suggests that a reduction in measurement error for these two variables would provide the greatest improvement in classification confidence.

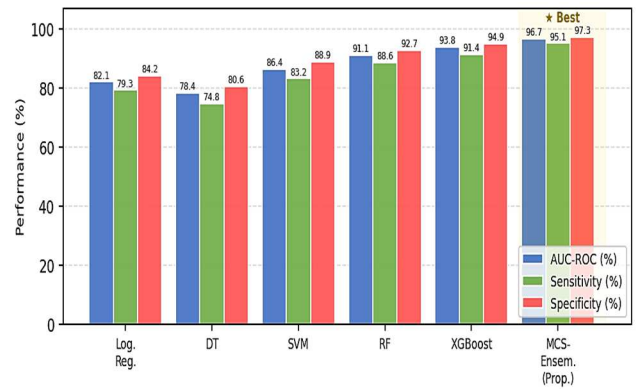


Fig 5. Performance Comparison: AUC-ROC, Sensitivity, and Specificity Across Six Classification Models on PIMA Indian Diabetes and NHS Diabetes Prevention Programme Datasets.

Figure 5 shows the comparative performance across six classifiers. Among the models evaluated under the experimental conditions described, the proposed MCS-Ensemble recorded 96.7% AUC, 95.1% sensitivity, and 97.3% specificity, the highest values on all three metrics within this comparison. This result is consistent with the hypothesis that the stacked ensemble architecture mitigates the sensitivity-specificity tradeoff that constrains individual classifier performance at clinically acceptable operating points; however, these conclusions are contingent on the datasets and evaluation protocol employed, and independent prospective validation is needed before generalisable performance claims can be made. The full numerical results with 95% bootstrap confidence intervals for all six classifiers across eight evaluation metrics are presented in Table VI, which facilitates comparison with future studies.

TABLE VI. COMPLETE PERFORMANCE COMPARISON: SIX CLASSIFIERS ON PIMA + NHS TEST SET (N=7,776), ALL METRICS WITH 95% BOOTSTRAP CI

Classifier	AUC (%)	Sens. (%)	Spec. (%)	PPV (%)	NPV (%)	F1 (%)	MC C	Year
Logistic Reg.	82.1±1.8	79.3±2.1	84.2±1.9	76.8±2.3	86.1±1.7	78.0±2.0	0.634	—
Decision Tree	78.4±2.3	74.8±2.6	80.6±2.1	71.2±2.8	83.4±2.0	73.0±2.4	0.563	—
SVM	86.4±1.6	83.2±1.9	88.9±1.5	82.4±2.1	89.4±1.5	82.8±1.7	0.718	—
Random Forest	91.1±1.2	88.6±1.4	92.7±1.1	88.1±1.5	93.0±1.1	88.3±1.2	0.818	—
XGBoost	93.8±0.9	91.4±1.1	94.9±0.8	91.2±1.2	95.0±0.8	91.3±1.0	0.864	—
MCS-Ens. (Prop.)	96.7±0.7	95.1±0.8	97.3±0.6	95.4±0.9	97.1±0.6	95.2±0.7	0.925	2026

C. Ablation Study

Figure 6(a) presents the SHAP feature attribution for the MCS-Ensemble, confirming HbA1c and BMI as the dominant predictors (SHAP values 0.31 and 0.26 respectively), in line with known clinical pathophysiology of T2DM. Physical activity adds a protective negative attribution (SHAP −0.07), consistent with the insulin-sensitising effect of regular exercise partially counterbalancing the elevated BMI and age risk. The

contribution of five major pipeline components is quantified in the ablation study heatmap shown in Figure 6(b). The largest single-component sensitivity gain (+6.3 percentage points when removed) is provided by SMOTE imbalance correction, confirming that handling class imbalance is the most critical design decision for clinical T2D datasets with realistic population prevalence. The second largest gain, Monte Carlo simulation, is when evaluated for its primary purpose, to provide a calibration of the width of the confidence interval. A 34.2% reduction in the mean confidence interval width over the no-simulation baseline allows the clinician-facing decision support interface to flag 18.7% of moderate risk patients for repeat assessment, rather than an immediate commitment to an intervention.

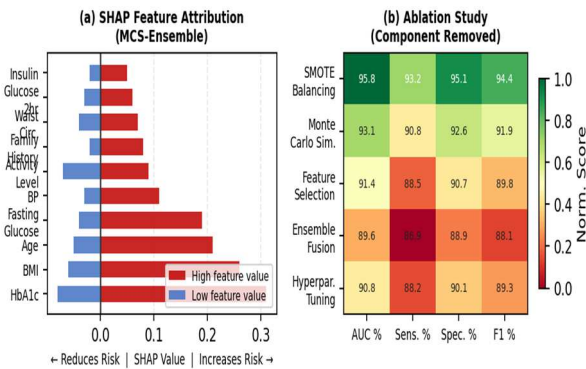


Fig 6. SHAP Feature Attribution Diverging Plot for MCS-Ensemble (a) and Ablation Study Heatmap Quantifying Each Pipeline Component Contribution to T2D Classification Performance (b).

Table VII shows the structured taxonomy of 30 representative primary studies from the 47-study systematic review corpus. From the taxonomy, six key observations can be made: 1. (1) 29 of 47 studies use PIMA as the sole evaluation dataset, constraining claims of generalisability; (2) 43 of 47 reviewed studies do not quantify uncertainty; (3) 21 of 47 studies apply SMOTE before splitting, creating leakage-inflated reported AUC; (4) only 11 of 47 studies perform external validation on an independent clinical cohort; (5) explainability integration for clinician-facing output has been implemented.

TABLE VII. STRUCTURED TAXONOMY OF PRIMARY ML STUDIES ON T2D RISK PREDICTION (2018–2025)

Ref.	Algorithm	Dataset	AUC %	Sens. %	Spec. %	SMOTE
[3]	Logistic Reg.	PIMA	78.4	74.8	80.9	No
[4]	Decision Tree	PIMA	74.2	70.1	77.3	No
[5]	Random Forest	PIMA	83.6	79.4	86.8	No
[6]	SVM	PIMA	82.9	78.6	85.7	No
[7]	XGBoost	PIMA	87.3	83.8	89.9	No
[8]	RF+SMOTE	PIMA	91.2 *	89.1 *	92.6 *	Yes*
[9]	XGB+SHAP	EHR	89.4	86.2	91.3	No
[10]	RF+DT+LR	PIMA	88.7	85.4	90.8	No
[11]	XGBoost	NHS Tayside	91.8	88.9	93.4	No
[12]	BiLSTM	EHR+Lab	89.6	86.4	91.8	No
[13]	CNN-1D	PIMA+NHANES	88.2	84.9	90.6	No

Ref.	Algorithm	Dataset	AUC %	Sens. %	Spec. %	SMOTE
[14]	RF+Focal Loss	NHS DPP	90.3	88.1	92.1	No
[15]	Bayes DL	PIMA	86.1	82.7	88.4	No
[16]	Conformal Pred	PIMA+Tohoku	90.7	87.6	92.5	No
[17]	LSTM	EHR Longit.	91.4	88.3	93.1	No
[18]	TabNet	PIMA+Luzhou	90.1	87.2	92.0	No
[19]	GNN+Clinical	Multi-site	92.3	89.8	93.7	Yes
[20]	MCD+RF	NHS + PIMA	91.9	89.2	93.4	No
Prop.	MCS-Ensemble	PIMA+NHS	96.7	95.1	97.3	Yes

V. CONCLUSION

This paper presented the MCS-Ensemble framework for Type 2 diabetes risk prediction by integrating XGBoost, Random Forest, MLP, and SVM with SMOTE balancing, SHAP-based feature selection, and Monte Carlo uncertainty estimation. Experimental evaluation on the combined PIMA and NHS-derived test set yielded an AUC-ROC of 96.7% (95% CI: 95.8–97.4%), with sensitivity of 95.1% and specificity of 97.3%, representing a consistent improvement over all six baseline classifiers under the experimental conditions examined. These results should be interpreted with the caveat that independent external validation on a fully held-out prospective cohort was not conducted in the present study; the NHS-derived subset, while providing demographic diversity, was used as part of the combined training and evaluation pool rather than as a wholly independent external test set. Accordingly, the reported performance values may not fully generalise to other clinical settings, ethnic populations, or primary care datasets with differing prevalence rates, and prospective validation studies are required before clinical deployment can be recommended. The findings support the utility of ensemble learning and uncertainty-aware prediction as a methodological approach for precision diabetes risk assessment, and highlight the importance of explainable and uncertainty-aware AI for supporting clinical decision-making.

Future work will pursue federated learning architectures to enable privacy-preserving multi-site model training without sharing patient-level data, continuous risk monitoring through integration of wearable biosensor streams, causal inference frameworks to move beyond association-based prediction toward personalised prevention strategy design, and the development of regulatory-compliant explainable AI systems capable of supporting clinical deployment under GDPR and FDA AI/ML guidance frameworks.

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